# Canada-Brazil Trade Analysis Report

## 2024-2025 Performance Review & 6-Month Forecast

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# Executive Summary

EXECUTIVE SUMMARY: CANADA-BRAZIL TRADE ANALYSIS (2024-2025)
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1. TRADE PERFORMANCE OVERVIEW
• Total trade changed by 16.1% ($2,005,093,576)
• Trade balance shifted by $-726,748,986
• SIGNIFICANT CHANGE: Trade volume changed by more than 10% between 2024 and 2025, indicating a major shift in bilateral trade relations.
• 8 provinces showed growth, 4 showed decline

2. TOP PERFORMERS
• Fastest growing province: Saskatchewan (47.0%)
• Top growing product: Special Classification Provisions II (408.6%)
• Key growth driver: Ontario (contributed 69.0% of total growth)

3. MARKET SHARE HIGHLIGHTS
• Largest market share (2025): Ontario (49.9%)
• Most improved market share: Ontario (+3.0 pp)
• Top 3 provinces control 92.4% of market

4. 12-MONTH OUTLOOK
• Expected growth rate: 2.9%
• Projected total trade: $7,698,085,039
• Year-over-year projection: 2.9% change
• Confidence level: MODERATE ($R^2$ = 0.428, MAPE = 8.9%)
• Model shows moderate predictive power - forecast has some

uncertainty

5. STRATEGIC RECOMMENDATIONS

- Focus on Saskatchewan as the primary growth region
- Prioritize Special Classification Provisions II category for expansion
- Monitor Saskatchewan for emerging opportunities
- Prepare for 2.9% trade volume increase over next 6 months

**16.1%**

TRADE VOLUME CHANGE

**$7,698,085,039**

6-MONTH PROJECTION

**$-726,748,986**

BALANCE SHIFT

**2.9%**

FORECAST GROWTH

# Q1: Trade Value & Balance Analysis (2024 vs 2025)

**$12,482,314,634**

TOTAL TRADE 2024

**$14,487,408,210**

TOTAL TRADE 2025

**$-7,981,143,768**

BALANCE 2024

**$-8,707,892,754**

BALANCE 2025

**Data Quality Note:** Chapter 98 (182 non-commercial records) excluded. 2,247 records ($322K) have UOM inconsistencies (0.001% of total value). All analyses use **trade VALUE (CAD $)** as primary metric.

**Significance Assessment:** SIGNIFICANT CHANGE - Trade volume changed by more than 10% between 2024 and 2025, indicating a major shift in bilateral trade relations.

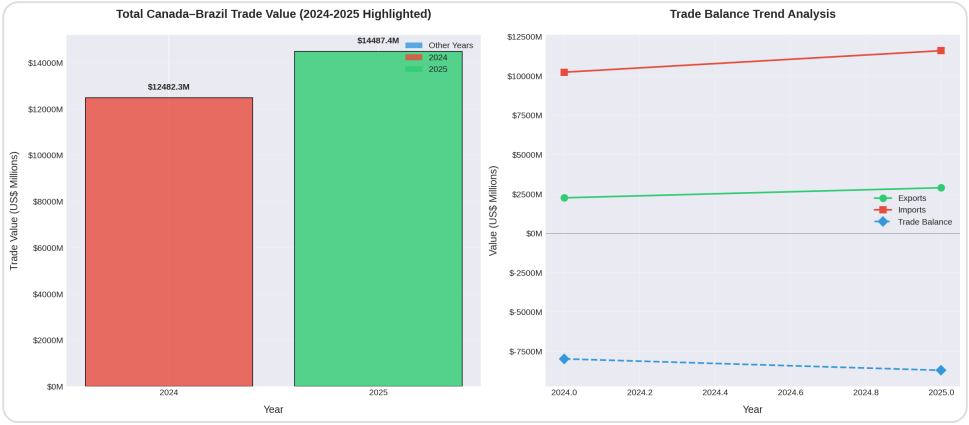


*Figure 1: Total Trade Value and Balance Trend (All Available Years)*

## Trade Statistics by Year

| Category | Year | Total Trade (Cad $) | Exports (Cad $) | Imports (Cad $) | Tr (C |
|---|---|---|---|---|---|
| 0 | $2,024 | $12,482,314,634 | $2,250,585,433 | $10,231,729,201 | $- |
| 1 | $2,025 | $14,487,408,210 | $2,889,757,728 | $11,597,650,482 | $- |

# Q2: Provincial Trade Performance

**8**

GROWING PROVINCES

**4**

DECLINING PROVINCES

**4.3%**

AVERAGE GROWTH

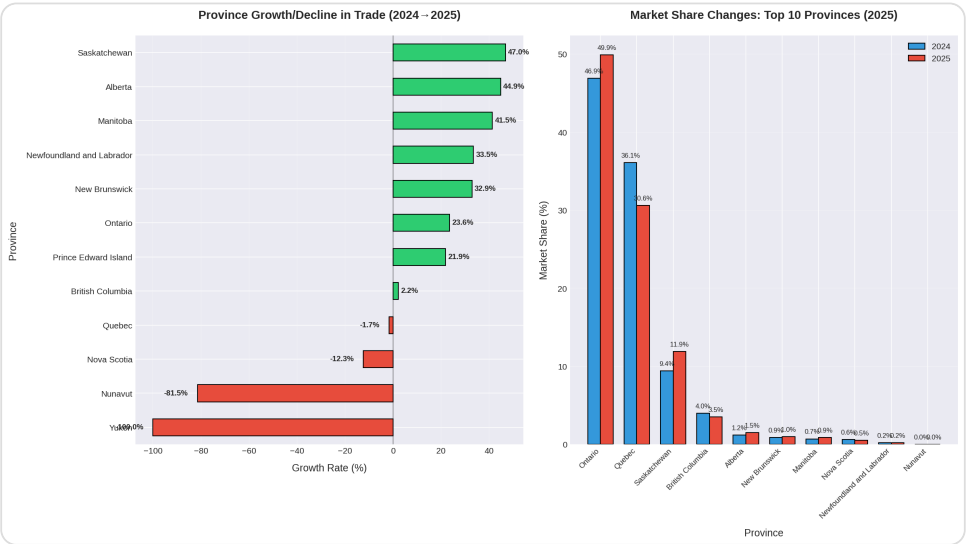


*Figure 2: Provincial Growth Rates and Market Share Changes (2024-2025)*

## Province Growth Analysis

| Category | Growth Rate (%) | Change ($) | Market Share 2025 (%) | Share Change (Pp) |
|---|---|---|---|---|
| Alberta | $45 | $68,222,093 | $2 | $0 |
| British Columbia | $2 | $11,217,789 | $4 | $-0 |
| Manitoba | $42 | $36,286,135 | $1 | $0 |
| New Brunswick | $33 | $35,751,672 | $1 | $0 |
| Newfoundland and Labrador | $34 | $7,691,500 | $0 | $0 |
| Nova Scotia | $-12 | $-9,201,278 | $0 | $-0 |
| Nunavut | $-82 | $-10,160 | $0 | $0 |
| Ontario | $24 | $1,382,820,305 | $50 | $3 |
| Prince Edward Island | $22 | $231,862 | $0 | $0 |
| Quebec | $-2 | $-76,219,142 | $31 | $-6 |
| Saskatchewan | $47 | $549,466,216 | $12 | $2 |
| Yukon | $-100 | $-1,163,416 | $0 | $0 |

# Q3: Comprehensive Trade Summary

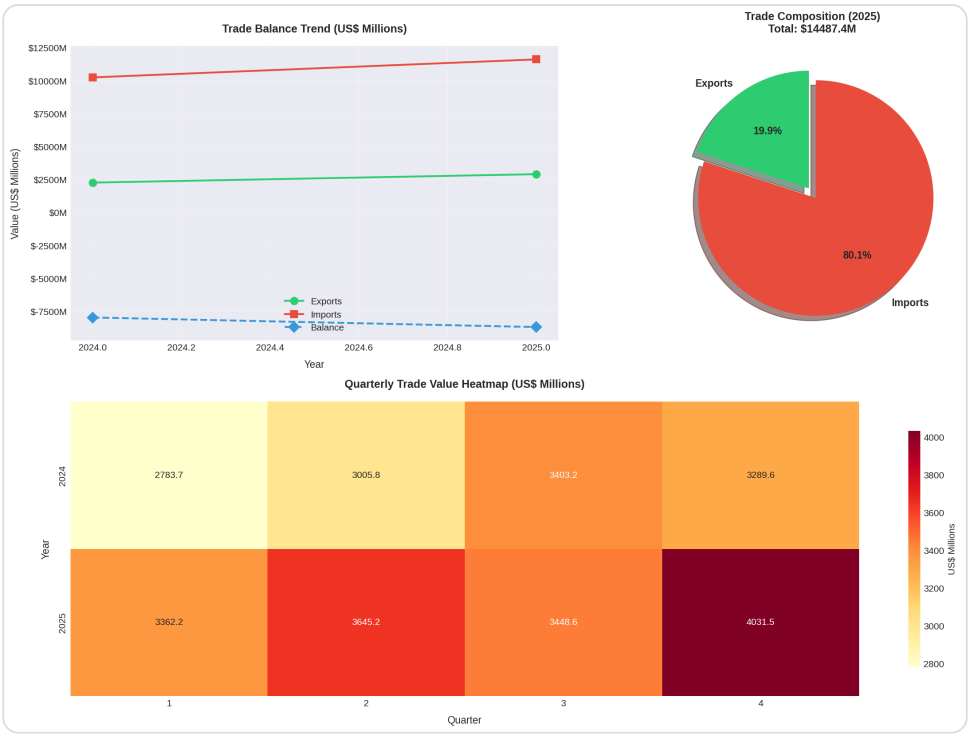


*Figure 3: Trade Balance Trend, Composition, and Quarterly Heatmap*

## Annual Trade Summary

| Category | Exports (Cad $) | Imports (Cad $) | Total Trade (Cad $) | Balance (Cad |
|---|---|---|---|---|
| **2024** | $2,250,585,433 | $10,231,729,201 | $12,482,314,634 | $-7,981,143 |
| **2025** | $2,889,757,728 | $11,597,650,482 | $14,487,408,210 | $-8,707,892 |

## Quarterly Trade Breakdown (CAD $)

| Category | Q1 (Cad $) | Q2 (Cad $) | Q3 (Cad $) | Q4 (Cad $) |
|---|---|---|---|---|
| **2024** | $2,783,670,733 | $3,005,848,625 | $3,403,238,852 | $3,289,556,42 |
| **2025** | $3,362,217,615 | $3,645,165,672 | $3,448,553,351 | $4,031,471,57 |

# Q4: Product Performance Analysis (HS Chapters)

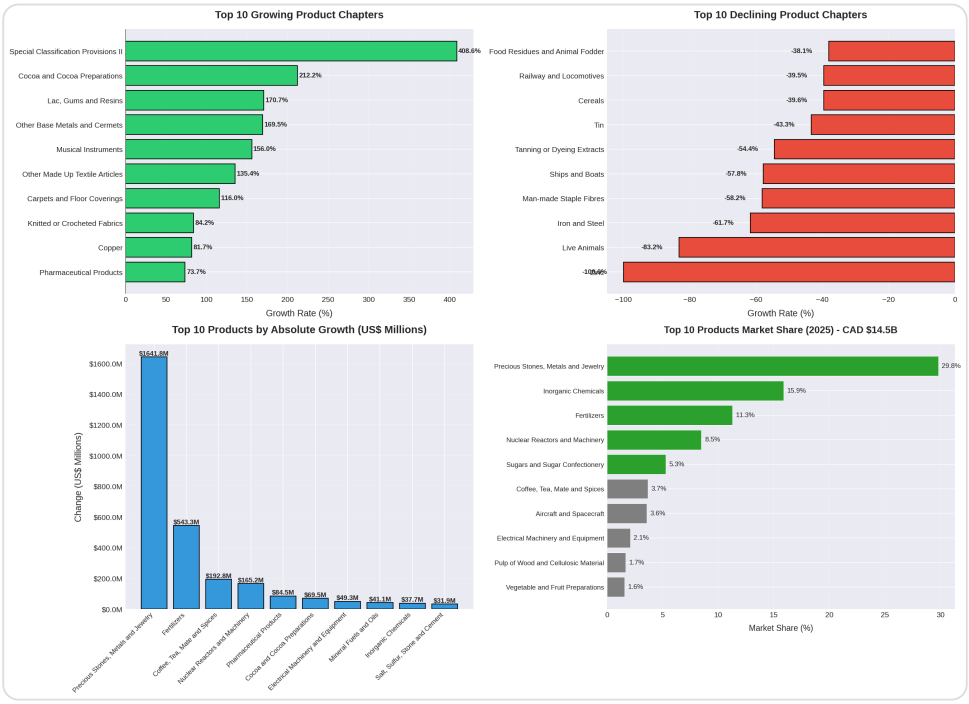


*Figure 4: Top Growing/Declining Products and Market Share Analysis*

## Top 10 Growing Products

| Category | Growth Rate (%) | Change ($) | 2024 Value ($) | 2025 Value ($) |
|---|---|---|---|---|
| Special Classification Provisions II | $409 | $12,443,127 | $3,045,379 | $15,488,506 |
| Cocoa and Cocoa Preparations | $212 | $69,455,974 | $32,723,918 | $102,179,892 |
| Lac, Gums and Resins | $171 | $5,277,431 | $3,092,217 | $8,369,648 |
| Other Base Metals and Cermets | $170 | $1,028,731 | $606,829 | $1,635,560 |
| Musical Instruments | $156 | $299,680 | $192,114 | $491,794 |
| Other Made Up Textile Articles | $135 | $1,288,931 | $952,054 | $2,240,985 |
| Carpets and Floor Coverings | $116 | $276,834 | $238,738 | $515,572 |
| Knitted or Crocheted Fabrics | $84 | $49,369 | $58,630 | $107,999 |
| Copper | $82 | $4,889,945 | $5,984,335 | $10,874,280 |
| Pharmaceutical Products | $74 | $84,521,766 | $114,724,759 | $199,246,525 |

## Top 10 Declining Products

| Category | Growth Rate (%) | Change ($) | 2024 Value ($) | 2025 Value ($) |
|---|---|---|---|---|
| Zinc | $-100 | $-132,926 | $132,972 | $46 |
| Live Animals | $-83 | $-745,998 | $896,317 | $150,319 |
| Iron and Steel | $-62 | $-296,421,965 | $480,695,479 | $184,273,514 |
| Man-made Staple Fibres | $-58 | $-121,716 | $209,284 | $87,568 |
| Ships and Boats | $-58 | $-869,059 | $1,503,921 | $634,862 |
| Tanning or Dyeing Extracts | $-54 | $-1,662,330 | $3,055,597 | $1,393,267 |
| Tin | $-43 | $-8,131,872 | $18,761,221 | $10,629,349 |
| Cereals | $-40 | $-13,768,049 | $34,777,907 | $21,009,858 |
| Railway and Locomotives | $-40 | $-1,377,491 | $3,484,609 | $2,107,118 |
| Food Residues and Animal Fodder | $-38 | $-6,218,844 | $16,338,792 | $10,119,948 |

# Q5: Province Contribution to Growth

**$2,005,093,576**

TOTAL GROWTH (2024-2025)

**69.0%**

TOP CONTRIBUTOR

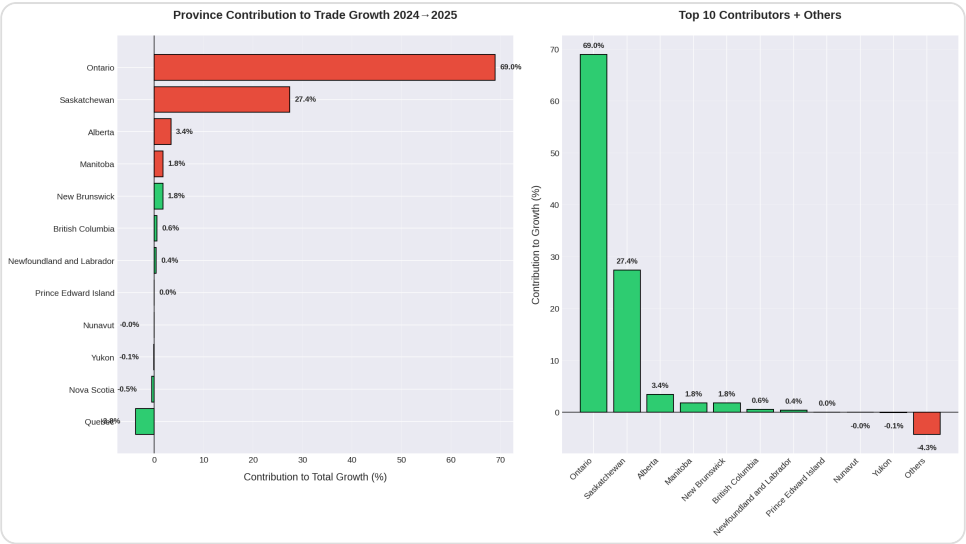


*Figure 5: Province Contribution to Total Trade Growth (2024-2025)*

## Province Contribution to Growth

| Category | Contribution (%) | Change ($) | Market Share 2025 (%) |
|---|---|---|---|
| Ontario | $69 | $1,382,820,305 | $50 |
| Saskatchewan | $27 | $549,466,216 | $12 |
| Alberta | $3 | $68,222,093 | $2 |
| Manitoba | $2 | $36,286,135 | $1 |
| New Brunswick | $2 | $35,751,672 | $1 |
| British Columbia | $1 | $11,217,789 | $4 |
| Newfoundland and Labrador | $0 | $7,691,500 | $0 |
| Prince Edward Island | $0 | $231,862 | $0 |
| Nunavut | $-0 | $-10,160 | $0 |
| Yukon | $-0 | $-1,163,416 | $0 |
| Nova Scotia | $-0 | $-9,201,278 | $0 |
| Quebec | $-4 | $-76,219,142 | $31 |

# Q6: 6-Month Forecast & Outlook

**2.9%**

EXPECTED GROWTH

**$7,698,085,039**

PROJECTED TOTAL

**MODERATE**

CONFIDENCE LEVEL

**0.428**

$R^2$ (MODEL FIT)

*Figure 6: Historical Data & 12-Month Forecast with Confidence Intervals*

## Forecast Model Comparison

### Model Comparison

| Category | Model | R2 | Mape (%) | Selected |
|---|---|---|---|---|
| **0** | Linear Regression (baseline) | 0.428 | 8.9 | |
| **1** | Prophet (Meta) | 0.965 | 2.4 | Yes |

## Forecast Details

- **Current 6-month average:** $1,246,670,820
- **Forecasted 6-month average:** $1,283,014,173
- **Year-over-year projected growth:** 2.9%
- **95% Confidence Interval:** ±$60,418,023 per month
- **80% Confidence Interval:** ±$39,271,715 per month
- **Model Fit ($R^2$):** 0.428

- **MAPE (Mean Absolute Percentage Error):** 8.9%
- **RMSE:** $119,969,263

  **Reliability Assessment:** Model shows moderate predictive power - forecast has some uncertainty

## Methodology & References

### Data Sources

- Statistics Canada - International Trade Database
- Brazilian Ministry of Economy - Comex Stat
- World Integrated Trade Solution (WITS)
- International Monetary Fund - Direction of Trade Statistics

### Methodology

- **Growth Rate Calculation:** ((2025 Value - 2024 Value) / 2024 Value) × 100
- **Forecast Method:** Prophet (Meta) multiplicative seasonality, 6-month horizon (80% and 95% confidence intervals)
- **Market Share:** (Provincial/Product Trade Value / Total Trade Value) × 100
- **Contribution Analysis:** (Provincial Change / Total Change) × 100
- **Pareto Analysis:** Identification of provinces contributing to 80% of growth
- **Seasonality:** Monthly average decomposition to identify cyclical patterns

### References


- World Trade Organization. "International Trade Statistics". Geneva: WTO Publications.
- Anderson, J.E. & van Wincoop, E. "Trade Costs and Trade Balance Dynamics". Journal of International Economics.
- Statistics Canada. "Canadian International Merchandise Trade Database". Ottawa: Statistics Canada.
- Brazilian Ministry of Economy. "Foreign Trade Statistics (Comex Stat)". Brasília: Ministry of Economy.


### Technical Notes

- All values are in Canadian Dollars (CAD), nominal terms (not adjusted for inflation)
- Data includes merchandise trade only (services excluded)
- HS Chapter classification based on Harmonized System 2022
- Forecast confidence intervals assume normal distribution of residuals
- Analysis period: 2024 - 2025
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